Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (EC) Examination

November 2019

EC 472 Radar Systems

Date: 13.11.2019, Wednesday **Time: 01:30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer following questions. **[07]**

(a) The frequency range for UHF band is___________.
(a) 30-300MHz (b) 300-3000MHz (c) 3-30MHz (d) 20-200MHz

(b) The term RADAR stands for_________.
(a) Radio direction and reflection (b) Radio dispatching and receiving
(c) Random detection and re-radiation (d) Radio detection and ranging

(c) An Altimeter is basically___________.
(a) a CW Radar (b) a FM Radar
(c) a Doppler Radar (d) a device to indicate the direction at height

(d) In DECCA ___________ phase difference is called a lane.
(a) 180 degree (b) 90 degree (c) 60 degree (d) 0 degree

(e) Each pulse group contains two sets of__________ pulse pairs spaced 100 μsec apart.
(a) 13-μsec (b) 11-μsec (c) 12-μsec (d) 10-μsec

(f) What is false alarm in radar?

(g) What is minimum detectable signal in radar?

Q - 2 Answer any two questions. **[14]**

(a) Derive basic radar equation.

(b) Explain important applications of radar in detail.

(c) Write a short note on four methods of navigation.

Q - 3 Answer any two questions. **[14]**

(a) Explain filter characteristics of Delay line canceller.

(b) Two MTI radars are found to have a blind speed are given by Vb2=20m/s of first MTI radar and second Radar at Vb3=30 m/s of second MTI radar if they operate on wavelength of 30 cm find the ratio of PRF of the Radar.

(c) An MTI radar is operating with 8Ghz with a PRF of 1200 Hz. Calculate the lower three blind speeds.

SECTION – II

Q-4 Answer following questions. **[07]**

(a) MTI radar operates on _______pulse repetition frequencies.
(a) Low (b) High (c) Constant (d) none

(b) ASR radar operates in _______ frequency Band.
(a) None (b) H band (c) Both (b) an (d) (d) E band

(c) Transponder is __________radar.
(a) Both (b) and (d) (b) Primary (c) none (d) Secondary

(d) CW radar sets transmit a _________frequency signal continuously.
(a) low (b) none (c) High (d) Constant

(e) What is Doppler principle?

(f) What is clutter in Radar?

(g) What is beat frequency in FMCW Radar?

Q-5 Answer following questions.

(a) Explain airborne radar. **[04]**

(b) Enlist various errors in direction finding. Explain any one of them in detail. **[05]**

(c) Describe the principle of VOR. Derive the equation of electric field of a VOR system with five antennas. **[05]**

OR

Q-5 Answer following questions.

(a) Explain synthetic aperture radar in detail. **[04]**

(b) Draw the block diagram of automatic Direction Finder. **[05]**

(c) Write a short note on VOR ground equipment with block diagram. **[05]**

Q-6 Answer following questions.

(a) What is navigation? Explain different types of navigation aids. **[05]**

(b) Write a short note on instrument landing system. **[05]**

OR

(b) Write a short note on microwave landing system. **[05]**

(c) Define direction finding term and which antenna is used for his purpose and why? **[04]**

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